

Lucian Mocan

lib.lucianmocan@gmail.com | linkedin.com/in/lucianmocan | lucianmocan.com

EXPERIENCE

Visiting Student Researcher Intern

Feb. 2026 – July 2026

Northwestern University

On-site, Evanston, Illinois, USA

- Researching sandboxed portable execution for Trusted Execution Environments in edge-computing settings
- Investigating ways to accelerate systems research using AI/LLMs

Language and Tooling Research Intern

June 2025 – Sep. 2025

University of Strasbourg

Remote, Knoxville, TN, USA

- Implemented **control-flow analysis** for functions and nested calls in **Althread's Rust** compiler and VM
- Built a modular **import system** by analyzing import mechanics in Promela, TLA+, Python, Go, and C++
- Developed **Althread's web IDE** (TypeScript, WASM) with a virtual filesystem, package manager, and debugger
- Authored comprehensive documentation and API guides for new users (writeup)

Full-stack Web Engineering Intern

June 2024 – Sep. 2024

Emerson

Cernay, France

- Translated purchasing workflows into an Angular + Express.js solution, **saving 800+ hours** of manual work
- Automated quote requests and centralized data management for the purchasing department (writeup)

EDUCATION

University of Strasbourg

Sep. 2024 – Aug. 2026

M.S. in Computer Science – Networks, Internet and Systems Engineering

Illkirch-Graffenstaden, France

- **Relevant Courses:** Advanced Algorithms, Compiler Design, Operating Systems Design, Inter- and Intra-Domain Routing, Cloud and Virtualisation, Distributed Algorithms, Embedded Programming, Advanced Cybersecurity

University of Strasbourg

Sep. 2021 – Aug. 2026

Cursus Master in Engineering (CMI): Internet, Systems and Networks

Strasbourg, France

- Selective 5-year track combining advanced coursework, research projects, internships, and international mobility.

University of Strasbourg

Sep. 2021 – June 2024

B.S. in Computer Science

Strasbourg, France

- **Honors:** Ranked 6th in class. **Courses:** Operating Systems Architecture, Algorithms & Data Structures, Software Engineering, Computer Networks, Artificial Intelligence, Parallel Computing

TECHNICAL SKILLS

Core: C/C++, Rust, TypeScript, WebAssembly, systems software, developer tooling

Systems: Linux, Bash, sockets, pthreads, MPI, OpenMP, shared memory, synchronization

Web/Infrastructure: Angular, Express.js, React Native, Docker, Git, SQL/MySQL

Networking: TCP/IP, routing protocols, DNS/DHCP, VLANs, P4

Languages: French, English, Romanian; Russian (intermediate)

PROJECTS

Sharing Square | *University of Strasbourg, Team Lead*

Sep. 2025 – Jan. 2026

- Led a 5-person team building a privacy-centric IoT platform for crowdsourcing health, environmental, and smart-city data for research (writeup)
- Built web/mobile components and integrated backend services, database infrastructure, and LLM-assisted data interaction features.

Althread Compiler | *University of Strasbourg (Solo Research)*

Feb. 2025 - May 2025

- Implemented **user-defined functions** in **Rust** for the Althread educational language (writeup)
- Extended the compiler's grammar, AST, and **stack-based VM** to support function calls, recursion, and nested expressions